

DEPARTMENT OF MECHANICAL ENGINEERING

Organizes

CHRONX-2026

RACING AGAINST TIME.

RULE BOOK-2026

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ORGANISATION AND GOVERNANCE

1. INTRODUCTION

1.1 About GCEE:

The college was established in 1984 as an Automobile Research Oriented Engineering college under Institute of Road and Transport by the Tamil Nadu State Transport Corporation. On 26 August 2021, Higher Education Minister of Tamil Nadu announced on Tamil Nadu Legislative Assembly that the IRTT was taken up by the government to directly run under the governance of Directorate of Technical Education (DoTE) and subsequently renamed as Government Engineering College, Erode. It is a premier Government Engineering College affiliated to Anna University, Chennai.. It is located in a sprawling lush green campus of 200 acres. It provides a congenial atmosphere for Technical Education. It is 15 km from Erode Railway Station and Half a km from the NH-47 between Bhavani and Chithode. The Institute offers 8 Under Graduate Degree courses in Engineering in various disciplines and 2 post graduate courses M.E Structural Engineering and M.E Computer Science & Engineering. The Institute has excellent infrastructure, well-equipped laboratories, library and highly qualified faculty members. The Institute is well known for its technical excellence, modern facilities, good performance track record and resourcefulness.

1.2. About the Department of Mechanical Engineering

At GCE, Erode, our students are well trained in all facets of Mechanical Engineering aided by state-of-the-art facilities. The Anna University curriculum consists of a balanced mix of theory and practice along with TN Government flagship skill development programme -Naan Mudhalvan. Also, our students are regularly undergo internship / in-plant training in prestigious industries like TAFE, TVS, Hyundai, Sakthi Auto Components Ltd. etc. Besides specialized laboratories like CAD/CAM/Mechatronics/IoT, our students have access to [Innovative Product Development Centre \(IPDC\)](#) to convert their innovative ideas to prototype / product.

1.3. About Racing Contest: **CHRONX**

CHRONX is the **first electric vehicle event ever conducted at GCEE**, marking a significant step forward in hands-on engineering initiatives. The event is designed to move beyond conventional symposium activities by providing students a platform to present real, functional electric vehicles and experience a structured, industry-style evaluation process focused on engineering quality rather than just speed.

Through CHRONX, students gain practical exposure to electric vehicle design, safety validation, efficiency, and real-world performance. The event helps participants understand how theoretical concepts translate into working systems, while also developing problem-solving skills, technical communication, and teamwork. As GCEE's inaugural EV event, CHRONX sets a strong foundation for future innovation and learning in electric mobility.

1.4. EVENT CATEGORIES:

SNO	VEHICLE CATEGORY.	MOTOR POWER.
1	ELECTRIC TWO WHEELER.	POWER<=1KW.
		POWER > 1KW.

1.5 COMPETITION OVERVIEW:

CHRONX is the EV Technical and Track challenge organised by Department of Mechanical Engineering, GCEE. This programme is designed to help students experience what real electric vehicle engineering looks like beyond classrooms and simulations. It gives participants an opportunity to convert their ideas into a working electric vehicle and understand how design decisions affect safety, performance, and efficiency in real conditions.

Through this competition, students develop **practical problem-solving skills**, learn to defend their engineering choices with logic, and gain confidence in handling real-world technical challenges. It encourages teams to think like engineers—balancing performance, reliability, energy usage, and safety rather than focusing on speed alone

It is a forum to showcase the talents of the students as a team comprising maximum of 5 student members. Teams are required to present a pre-built electric vehicle that has been designed and fabricated in accordance with the criteria specified in the rulebook. It is the responsibility of each team to ensure that their vehicle fully complies with all prescribed requirements and is ready for demonstration during the event.

This event provides maximum flexibility for the participants to bring forth their creativity and imagination but there is certain restriction for the vehicle design based on the category they fall in.

1.6 EVENT DATE AND VENUE:

The event takes place at **Government College of Engineering, Erode – 638316.**

The event will be held on **5th March, 2026.**

2. REGISTRATION:

The maximum number of team members is 5 (Five).

All the correspondence should take place from the Team's official mail ID.

Registration fees for each event category is mentioned below and it will not be refunded in any case.

- **E-Bike Category** – Rs. 4000/- (Four Thousand only).
(For teams exceeding five members, an additional registration fee of ₹300 per extra member shall be applicable.).

The registration for the event will be closed by **1st March 2026**.

2.1 Post registration process:

Once the registration is completed, the vehicle will be given a number and that vehicle will be known by that number throughout the whole event.

Any program related official news regarding the event will be informed to the team via team official mail id (Corresponding mail id).

3. SAFETY REQUIREMENTS:

- 1) Helmet, gloves, shoes, elbow pads and knee pads must be there for Rider.
- 2) The driver seat must be fixed rigidly.
- 3) Each team should have two drivers and they must have a valid driver license (learner's license is not allowed) and it must be shown to the inspection team.
- 4) For two-wheeler category, drivers must have goggles or visor in the helmet.
- 5) Battery, battery management system (optional) and motor should be provided with rain protection to avoid entry of rain shower and splash from water on road surface.
- 6) Every vehicle must have brake light.

4. TECHNICAL INSPECTION:

The vehicle must be presented in front of the technical inspection committee jury in fully finished condition (ready to run condition).

The vehicle will be re-inspected if any correction given by the TI committee.

After completion of technical inspection (TI), no modification or adjustment should be made without the permission of the TI committee. If any teams violating the above will be disqualified.

Any decision taken by the TI committee will be final and they will be given full authority regarding the selection of teams for the dynamic events.

4.1 OFFICIAL LANGUAGE:

The official language of the CHRONX-2026 is English. Document submissions, presentations and discussions should be done in English

5. PRIMARY RULES:

5.1 Material and Cross section required:

Any material of minimum 0.1% carbon can be used. A circular cross section of minimum diameter 1 inch and 1.65 mm thickness is recommended. Usage of any other cross section requires proper analysis report. The material and cross section must be chosen in such a way that it can withstand collisions during dynamic events.

5.2 Kill switch:

Push-to-off kill switch must be provided on the vehicle. Whole electrical circuit of drivetrain must get dead by pushing off the kill switch. **AT LEAST ONE** kill switch must be easily accessible to the driver.

5.3 Protection for the batteries:

Proper shielding should be provided to protect batteries from water splash, dust and mud etc. In all cases the safety of CHRONXrs must be ensured. Short-circuit/fire/explosion prevention techniques should be applied. Soft packing of batteries is prohibited, only hard packing is allowed. Batteries should be mounted on a rigid frame attached to the vehicle frame. Hung mountings are not allowed. The mounting should be able to protect batteries from falling at the time of bumps and leaning etc.

A. **Protection for the driver and passengers:** All those moving parts must be shielded to prevent injury to the driver and passenger(s). All electrical components must be properly shielded from the driver. To avoid any short circuit, battery terminals must be shielded but should be kept accessible for the approach of any measuring instrument. All electrical connections should be properly insulated from the frame.

All wires and harnesses must be attached securely to the vehicle structure that prevents coming off in static and dynamic conditions. Use of metal wires, synthetic threads and tapes as a fastening device is prohibited. The wires and cables must be routed along the frame in a flexible casing and should be tied to frame such that these do not entangle with the rider's body and other moving parts of the vehicle.

6 TECHNICAL SPECIFICATIONS REQUIRED:

6.A.1 BASIC REQUIREMENTS:

- The wheelbase of the bike should be in the range of 1150 mm to 1350mm(48 to 62 inches).
- The maximum length should not exceed more than 1800mm (70 inches).
- The maximum width of the bike should not be more than 750mm (40 inches).
- The maximum height of bike without rider should not be more than 1200mm (60 inches).
- The weight of the bike must not exceed 150kg and minimum weight of bike without rider is 35kg.
- The minimum ground clearance should be 150 mm(5 inches).
- The bike can be a single seater or double seater.
- The wheel size must between 10 inches to 12 inches.
- The maximum allowed width of the tire is 3 to 3.5 inches.
- Bicycle tires are prohibited.
- The use of front and rear mudguards is compulsory.

6.A.2 FRAME DESIGN:

Teams can use either Retrofitting Class or Self-Manufactured class but they should obey all the above mentioned specifications. Proper mounting for the battery and motor should be there.

6.A.3 BRAKE AND SUSPENSION SYSTEM:

- A hydraulic or non-hydraulic braking system must be there, and it must actuate properly during the brake test.
- A suspension system which actuates properly during dynamic events is mandatory.

6.A.4 BATTERY AND MOTOR SPECIFICATIONS:

Teams are allowed to use Lead Acid or Li-Ion batteries. Team will not be allowed to change the battery pack. They need to use the same battery pack throughout the event.

Allowed Nominal Voltage will be from 48 V to 72V.

Energy Allowed :

Minimum Energy: 1.0KWh

Maximum Energy: 2.5 KWh.

Teams are allowed to use any type of motor, AC or DC, and it should vary from 500W to 2KW.

6.A.5 TRANSMISSION SYSTEM:

Use of chain, CVT or belt drive is permitted for delivering power. Direct transmission between the output shaft of the motor and the rear wheel is allowed.

7 EVENTS:

The events are categorized into two categories, that is

- i. Static event- 600 points.
- ii. Dynamic events- 400 points.

Note: The scores for these events will be awarded **category wise**.

A. STATIC EVENTS:

7.A.1 TECHNICAL INSPECTION (TI):

The objective of Technical Inspection round is to determine whether the ability of the vehicle meets the requirements and restrictions of rules. Technical inspection doesn't have any points, but every vehicle must pass all parts of Technical Inspection and testing before it is permitted to participate in any dynamic event.

The exact procedures and instruments employed for inspection and testing are entirely at the discretion of the Chief Technical Inspector. vehicle must be presented for technical inspection in finished condition, i.e., fully assembled, complete and ready-to-run. Technical inspectors will not inspect any vehicle presented for inspection in an unfinished state.

Corrections: If a vehicle does not comply with the rules, then correction must be done to be included in re-inspection process.

The lab report of the material and analysis reports of the frame must be submitted for the verification of the technical inspector. Weight check of the vehicle will be done by the technical inspectors.

7.A.2 DESIGN AND ENGINEERING DEFENCE:

The teams must explain and justify their vehicle design and engineering decisions before the judges. Evaluation is based on clarity of design objectives, component selection rationale, supporting calculations, safety considerations, manufacturability, and cost effectiveness. Teams are expected to demonstrate a thorough understanding of their vehicle architecture and clearly articulate the trade-offs made during design and fabrication.

7.A.3 GENERAL MCQ:

For the General MCQ Test on Electric Vehicles, a question bank consisting of 50 multiple-choice questions will be prepared by the organizing committee, covering fundamental and applied concepts related to electric vehicles. Out of these 50 questions, 30 questions will be selected for the actual event. The selection of questions may be random or as decided by the organizers to ensure fairness and uniformity across all participating teams. The MCQ test shall be common for all teams and will be conducted as part of the static events. Evaluation will be based solely on the correctness of responses, and the scores obtained in this test will contribute to the overall static event score. The decision of the organizing committee regarding question selection and evaluation shall be final and binding.

7.A.4 FAILURE SCENARIOS AND ENGINEERING RESPONSE:

The Failure Scenarios and Engineering Response event evaluates the team's ability to handle real-world engineering challenges. Teams are presented with potential failure situations related to electric vehicles and are assessed on their ability to identify root causes, evaluate safety risks, propose immediate corrective actions, and suggest long-term design or procedural improvements. This event emphasizes logical reasoning, troubleshooting skills, and practical engineering judgment.

B. POINTS DISTRIBUTION OF STATIC EVENTS:

EVENTS	POINTS
TECHNICAL INSPECTION	200
DESIGN AND ENGINEERING DEFENCE	150
GENERAL MCQ's	100
FAILURE SCENARIOS AND RESPONSE	150
TOTAL	600

C. DYNAMIC EVENTS:

7.C.1 BRAKETEST:

The vehicle will be allowed to run on a 75 m long track, on which it must attain a MAXIMUM speed of 35kmph. Then, it will be allowed to brake after crossing a certain line called braking line and it must stop within 6m after crossing braking line. Only after clearing the braking test, the vehicle will be allowed to attend other dynamic events. Braking test does not have any points.

7.C.2 ACCELERATION TEST:

During this test, the vehicle (except two-wheeler category) must carry payload along with the driver (no passenger). The vehicle needs to travel 85m long track with its maximum acceleration. According to the time taken, points will be granted. The team with minimum time gets maximum score.

7.C.3 CONSTANT SPEED LAP TEST:

The Constant Speed Lap Test checks how well an electric vehicle performs when driven calmly and consistently, just like in real-world use. In this test, teams must complete a set number of laps while maintaining a steady speed within the 35 km/h limit. The goal is not to go fast, but to show smooth control, stable performance, and efficient use of energy throughout the run.

___This test highlights how well the motor, controller, battery, and drivetrain work together under normal operating conditions. Vehicles that can maintain a steady speed with minimal energy loss demonstrate better engineering, tuning, and system reliability. Any sudden speed changes, unsafe riding, or violation of rules may lead to penalties, as the test emphasizes control, efficiency, and endurance rather than racing.

PENALTIES:

- For each cone the vehicle takes down, a penalty of 3 seconds will be added.
- If the vehicle leaves the track, it must re-enter at the same point or prior to it or 15 seconds penalty will be given.
- For two-wheelers, leg touching the ground gives a penalty of 3 seconds.
- If any other mistakes is done by the driver, appropriate penalties will be imposed.

D. Points distribution of dynamic events:

EVENTS	POINTS
Braketest	100
Acceleration Test	100
Lap test	200
TOTAL	400

8 EVENT RULES AND ORGANIZER AUTHORITY:**8.1 RULES COMPLIANCE:**

By entering CHRONX, the team members & faculty advisors of the entering university agree to comply with, and be bound by, the rules of CHRONX-2026. All the rules & procedures are issued or announced by the CHRONX-2026 Organizing Committee. All team members, faculty advisors and other university/college representatives are requested to cooperate with us and follow all the instructions from competition organizers, officials and judges.

8.2 BEHAVIOUR:

All the members of every team are required to follow the rules laid by CHRONX 2024 Organizing Committee, during the competition. Any team fails to follow the rules will result in elimination from the event. Arguments with officials may also result in the team being eliminated from event.

In the event of unsportsmanlike conduct, the team will receive a warning from an official. A second violation will result in expulsion of the team from the competition.

Alcohol, illegal drugs, weapons or other illegal material are strictly not allowed on the event site during the competition. Any violation of this rule by a team member will cause the expulsion of the entire team. Any use of drugs, or the use of alcohol by an underage individual, will be reported to the authorities for prosecution.

8.3 OFFICIAL INSTRUCTIONS:

Failure of a team member(s) to follow an instruction or command directed specifically to that team or team member(s) will result in a thirty-point penalty. There should not be any direct involvement of industrial experts in Designing and manufacturing of the vehicle.